

SOLIDWORKS API (.NET VBA C#) — Reference (195/337): IModelDocExtension (part 1 of 8)

SOLIDWORKS API Documentation (converted from CHM)

IModelDocExtension (part 1 of 8) (Type)

IModelDocExtension Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IModelDocExtension Interface

Allows access to the model.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IModelDocExtension
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension
```

C#

```
public interface IModelDocExtension
```

C++/CLI

```
public interface class IModelDocExtension
```

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension](#).

Example

- [Access Assembly \(C++\)](#)
- [Create Cylinder \(C++\)](#)
- [Traverse Bodies \(C++\)](#)
- [Get Names of Bodies in Multibody Part \(VBA\)](#)
- [Insert Wrap Feature \(VBA\)](#)
- [Use Persistent Reference \(VBA\)](#)
- [Create Equation-driven Curve \(C#\)](#)
- [Create Equation-driven Curve \(VB.NET\)](#)
- [Create Equation-driven Curve \(VBA\)](#)

Accessors

[IModelDoc2::Extension](#)

Access Diagram

[ModelDocExtension](#)

See Also

- [IModelDocExtension Members](#)
- [SolidWorks.Interop.sldworks Namespace](#)

IModelDocExtension Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IModelDocExtension Interface

The following tables list the members exposed by [IModelDocExtension](#).

Public Properties

Name	Description
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 ActiveCommandTab	Gets and sets the active SOLIDWORKS CommandManager tab.
 ActiveCommandTabIndex	Gets and sets the index of the active SOLIDWORKS CommandManager tab.
 AnnotationViewCount	Gets the number of annotation views in this part or assembly document.
 AnnotationViews	Gets the annotation views in this part or assembly document.
 AppPageSetup	Gets the SOLIDWORKS Design application page setup interface for this document.
 CommandTabVisible	Gets and sets the visibility of the specified SOLIDWORKS CommandManager tab.
 CustomPropertyBuilderTemplate	Gets or sets the custom property builder template for this part.
 CustomPropertyManager	Gets the custom properties for this document or configuration.
 DimXpertManager	Gets the DimXpert schema for this configuration.
 DisplayMode	Gets and sets the display modes for the specified display state setting.
 DisplayStateSpecMaterialPropertyValues	Gets and sets the appearance settings for the specified display state setting.
 DisplayTitle	Gets and sets this model's title to display in the FeatureManager design tree.
 Document	Gets the model document.
 FeatureManagerFilterString	Gets or sets the string in the FeatureManager design tree filter.
 FlyoutFeatureTreeVisibility	Gets or sets the state of the flyout FeatureManager design tree.
 HasLegacyCustomProperties	Gets whether this model has legacy custom properties.
 IncludeMassPropertiesOfHiddenBodies	Gets or sets whether to include the mass properties of hidden components in the assembly.
 LinkedDisplayState	Gets or sets whether a display state is linked in this part.
 NeedsRebuild	Obsolete. Superseded by IModelDocExtension::NeedsRebuild2.
 NeedsRebuild2	Gets whether the model document needs to be rebuilt.
 ShowPartRebuildIndicators	Gets or sets whether to display rebuild indicators on parts that have out-of-date frozen features.
 SunLightInformation	Gets the specified sunlight information.
 ToolboxPartType	Gets and sets whether this part is a SOLIDWORKS Toolbox part.
 Transparency	Gets and sets the transparency states for the specified display state setting.
 UsePageSetup	Gets or sets whether this document uses its own page setup values, SOLIDWORKS Design application page setup values, or setup values on individual drawing sheets.
 ViewDisplayRealView	Gets or sets the RealView Graphics setting.
 Visibility	Gets and sets the visibility states for the specified display state setting.

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Public Methods

Name	Description
 ActivateFeatureMgrViewx64	Activates the specified FeatureManager view in x64 applications.
 AddAngularRunningDim	Adds the specified angular running dimension for selected entities.
 AddComment	Adds a comment to this document's Comment Folder.
 AddDecal	Adds a decal to the model.
 AddDefaultRenderMaterial	Not supported in SOLIDWORKS 2011 and later and not superseded.
 AddDimension	Creates a display dimension at the specified location for selected entities.
 AddDisplayStateSpecificRenderMaterial	Adds the specified appearance to the specified display states in the active configuration and returns the IDs assigned to that appearance.
 AddFeatureMgrView3x64	Adds the specified tab to the FeatureManager design tree view in x64 applications.
 AddOrdinateDimension	Inserts an ordinate dimension.
 AddOrUpdateSearchData	Adds or updates the SOLIDWORKS Search, third-party, application keyword and value to the model document.
 AddPathLengthDim	Inserts a path length dimension at the specified coordinates for a selected path.
 AddRenderMaterial	Not supported in SOLIDWORKS 2011 and later. Superseded by IModelDocExtension::AddDisplayStateSpecificRenderMaterial and IModelDocExtension::IAddDisplayStateSpecificRenderMaterial.

	AddSpecificDimension	Creates the specified display dimension at the specified location for selected entities.
	AddSymmetricDimension	Creates a full symmetrical angular dimension at the specified location for the selected entities.
	AlignDimensions	Aligns the selected dimensions in drawing documents.
	AlignRunningDimension	Aligns the extension lines of all angular dimensions to be the same distance from the center as the baseline dimension (0°) in the set of angular running dimensions.
	ApplyFormatPainterToAll	Applies Format Painter to all dimensions and annotations in the active document.
	ApplyTexture	Obsolete. Superseded by !ModelDocExtension::SetTexture .
	BreakAllExternalFileReferences2	Breaks all external references and allows you to insert the features of the original part, or parts, if the external references are broken.
	Capture3DView	Captures the 3D View of this part or assembly.
	ChangeSketchPlane	Changes the plane used by a sketch by moving the selected sketch to the selected plane in the specified configurations.
	CheckVersionCompatibility	Checks whether a document can be saved in the file format for an earlier version of SOLIDWORKS.
	CloseMessageBar	Closes the specified message bar in this document's window.
	CloseUserNotification	Closes the specified user notification in this document's window.
	Compare3DPMI	Compare DimXpert annotations, reference dimensions, and other annotations between different versions of the same part document.
	CopyDraftingStandard	Copy the current custom drafting standard.
	Create3DBoundingBox	Creates a 3D bounding box for a cut list item in a weldment part.
	CreateAdvancedHoleElementData	Creates an Advanced Hole element data object of the specified type.
	CreateBalloonOptions	Creates an object that stores BOM balloon options.
	CreateCallout	Creates a callout independent of a selection.
	CreateDecal	Creates a decal for this model.
	CreateMassProperty	Obsolete. Superseded by !ModelDocExtension::CreateMassProperty2 .
	CreateMassProperty2	Creates a mass properties object.
	CreateMeasure	Creates a measure tool.
	CreateOLEObject	Creates an OLE object on the active document.
	CreateRenderMaterial	Creates an appearance for this model.
	CreateStackedBalloonOptions	Creates an object that stores options for stacked balloons.
	CreateStructureSystem	Creates a structure system feature using the specified primary and secondary member arrays.
	CreateStructureSystemMemberData	Creates the specified structure system member.
	CreateTexture	Creates a texture.
	DeActivateFeatureMgrViewx64	Deactivates a tab in the FeatureManager design tree view in x64 applications.
	DeleteAllDecals	Deletes all decals on this model.
	DeleteAttachment	Deletes the specified file in the Attachments folder in the FeatureManager design tree.
	DeleteDecal	Removes the specified decal from this model.
	DeleteDisplayStateSpecificRenderMaterial	Deletes the specified appearances, using the IDs of the appearances, from the active configuration.
	DeleteDraftingStandard	Delete the current custom drafting standard.
	DeleteFeatureMgrViewx64	Removes the specified tab in the FeatureManager design tree in 64-bit applications.
	DeleteRenderMaterial	Not supported in SOLIDWORKS 2011 and later. Superseded by !ModelDocExtension::DeleteDisplayStateSpecificRenderMaterial and !ModelDocExtension::!DeleteDisplayStateSpecificRenderMaterial .
	DeleteScene	Deletes the scene applied to this model.
	DeleteSearchData	Deletes the specified SOLIDWORKS Search third-party keywords from the model document.
	DeleteSelection2	Deletes the selected items, with the option to delete absorbed features, child features, or both.
	EditBalloonProperties	Obsolete. Superseded by !ModelDocExtension::EditBalloonProperties2 .
	EditBalloonProperties2	Edits the selected balloon's properties.
	EditDimensionProperties	Edits the selected dimension.
	EditRebuildAll	Rebuilds only those features that need to be rebuilt in all configurations without activating each configuration.

		
	FindTrackedObjects	Finds the tracking IDs assigned to entities in this document.
	FinishRecordingUndoObject	Obsolete. Superseded by IModelDocExtension.FinishRecordingUndoObject2 .
	FinishRecordingUndoObject2	Ends recording of a SOLIDWORKS Undo object with the specified name and visibility.
	ForceRebuildAll	Forces a rebuild of all features in all configurations without activating each configuration.
	GeodesicSketchOffset	Creates a geodesic sketch offset along the curvature of a surface.
	Get3DEXperienceModelType	Gets the type of 3DEXPERIENCE application that owns this model.
	Get3DView	Gets the 3D View with the specified name for this part or assembly.
	Get3DViewCount	Gets the number of 3D Views in this part or assembly.
	Get3DViewNames	Gets names of the 3D Views in this part or assembly.
	Get3DViews	Gets the 3D Views for this part or assembly.
	GetActiveAnnotationView	Gets the active annotation view of this model.
	GetActivePropertyManagerPage	Gets the name of the active PropertyManager page.
	GetAdvancedSaveAsOptions	Gets the advanced File > Save As options.
	GetAdvancedSpotLightProperties	Gets the attenuation-related, advanced properties for the specified SOLIDWORKS spot light in this model.
	GetAnnotationCount	Gets the number of annotations on this part.
	GetAnnotations	Gets the annotations on this part.
	GetAnnotationsByType	Gets the annotations of the specified type in this document.
	GetAppearanceSetting	Gets the appearance setting for this document.
	GetAttachmentCount	Gets the number of attachments for this document.
	GetAttachments	Gets the attachments for this document.
	GetCalloutVariableString	Gets the string for the specified callout variable.
	GetCameraById	Gets a camera using the specified camera ID.
	GetCameraCount	Gets the number of cameras in the document.
	GetCameraDefinition	Gets a camera, but does not add the newly created camera to the model.
	GetCommandTabs	Gets all of the SOLIDWORKS CommandManager tab names in this document.
	GetCoordinateSystemTransformByName	Gets the transform of the specified coordinate system.
	GetCorresponding	Obsolete. Superseded by IModelDocExtension.GetCorresponding2 .
	GetCorresponding2	Gets the object in the underlying part or subassembly document that corresponds to the specified input object in this drawing view or assembly.
	GetCorrespondingEntity	Obsolete. Superseded by IModelDocExtension.GetCorrespondingEntity2 .
	GetCorrespondingEntity2	Gets the entity in the underlying part or subassembly that corresponds to the specified entity in this assembly or drawing view.
	GetCostingManager	Gets the entry-point interface to the SOLIDWORKS Costing API and gets the CostingManager.
	GetCSYSDistances	Gets the specified distances between two selected coordinate systems.
	GetCSYSEulersAngularRotation	Gets the specified Eulers angular rotation values that transform one selected coordinate system into another.
	GetCSYSXYZAngularRotation	Gets the specified Tait-Bryan angular rotation values that transform one selected coordinate system into another.
	GetDecal	Gets the specified decal in this model.
	GetDecals	Gets the decals applied to the model.
	GetDecalsCount	Gets the number of decals applied to this model.
	GetDependencies	Gets all of this model's dependencies.
	GetDisplayStateSetting	Gets the display state setting for the specified scope.
	GetDraftingStandardNames	Get the names of all currently available drafting standards.
	GetFlattenSheetMetalPersistReference	Gets a byte array of persistent reference IDs for the specified entity (a face, edge, or vertex) in a flattened sheet metal part.
	GetGeneralTableAnnotation	Creates a general table annotation for SOLIDWORKS MBD 3D PDF.

		
	GetKeepLightInRenderScene	Gets whether a light is kept when the scene changes.
	GetLastFeatureAdded	Gets the last feature added to the model.
	GetLicenseType	Gets the type of SOLIDWORKS license used when the model was created.
	GetLightEnabledInRender	Gets whether a light is on in this model.
	GetMassProperties	Obsolete. Superseded by IModelDocExtension::GetMassProperties2 .
	GetMassProperties2	Obsolete. Superseded by IMassProperty2 .
	GetMaterial	Gets the appearance for the specified appearance ID in the specified configuration in this model document
	GetMaterialPropertyValues	Gets the material properties for this model document.
	GetMBD3DPdfData	Gets the SOLIDWORKS MBD 3D PDF data object.
	GetMBDSTEP242Data	Gets the SOLIDWORKS MBD STEP 242 data object.
	GetModelBreakViewNames	Gets the names and number of the Model Break Views in the current configuration of the active model.
	GetModelViewCount	Gets the number of model views in this document.
	GetModelViews	Gets the model views in this document.
	GetMotionStudyManager	Gets the SOLIDWORKS motion study's MotionManager.
	GetNamedViewRotation	Gets the specified named view orientation matrix with respect to the Front view.
	GetObjectByPersistReference	Obsolete. Superseded by IModelDocExtension::GetObjectByPersistReference3 .
	GetObjectByPersistReference2	Obsolete. Superseded by IModelDocExtension::GetObjectByPersistReference3 .
	GetObjectByPersistReference3	Gets the object assigned to the specified persistent reference ID.
	GetObjectId	Gets the object ID for the specified annotation.
	GetOLEObjectCount	Gets the number of OLE objects.
	GetOLEObjects	Get the OLE objects.
	GetPackAndGo	Gets a Pack and Go object.
	GetPersistReference	Obsolete. Superseded by IModelDocExtension::GetPersistReference3 .
	GetPersistReference3	Gets the persistent reference ID for the specified object in this model document.
	GetPersistReferenceCount	Obsolete. Superseded by IModelDocExtension::GetPersistReferenceCount3 .
	GetPersistReferenceCount3	Gets the size of the persistent reference ID assigned to the selected object in this model document.
	GetPLMID	Gets the ID of this SOLIDWORKS Design connected to the 3DEXPERIENCE platform document.
	GetPrint3DDialog	Gets the IPrint3DDialog object.
	GetPrintSpecification	Gets the IPrintSpecification object for this document.
	GetRenderCustomReferences	Get the custom render references for this model.
	GetRenderMaterials	Obsolete. Superseded by IModelDocExtension::GetRenderMaterials2 .
	GetRenderMaterials2	Gets the appearances applied to this model document in the specified display states.
	GetRenderMaterialsCount	Obsolete. Superseded by IModelDocExtension::GetRenderMaterialsCount2 .
	GetRenderMaterialsCount2	Gets the number of appearances applied to this model document in the specified display states.
	GetRenderStockReferences	Gets the SOLIDWORKS-supplied (stock) render references for this model.
	GetRoutingComponentManager	Gets the routing component manager.
	GetScanto3D	Gets the IScanTo3D object for this document.
	GetSceneBkgDIBx64	Gets the background image as DIBSECTION in 64-bit applications.
	GetSearchData	Gets the SOLIDWORKS Search, third-party, application keywords from the model document.
	GetSearchDataCount	Gets the number of SOLIDWORKS Search keywords for the specified third-party application previously added to this model document.
	GetSectionProperties	Obsolete. Superseded by IModelDocExtension::GetSectionProperties2 .
	GetSectionProperties2	Gets the section properties for the following types of selected items:



- Planar model face in any document
- Face on a section plane
- Crosshatch section face in a section view in a drawing a sketch
- Sketch



[GetSheetMetalObjectsByPersistReference](#)

Gets the objects in a folded sheet metal part that correspond to the byte array of persistent reference IDs of an entity in a flattened sheet metal part.



[GetSphericalBoxDiameter](#)

Gets the diameter of the spherical bounding box of this model.



[GetStream](#)

Gets the handle for the specified stream.



[GetSunLightAdvancedPropertyValues](#)

Gets the specified sunlight advanced properties.



[GetSunLightSourcePropertyValues](#)

Gets the property values for a sunlight source.



[GetSustainability](#)

Gets the entry-point interface to the SOLIDWORKS Sustainability API.



[GetTemplateSheetMetal](#)

Gets the sheet metal folder feature from this sheet metal model created in SOLIDWORKS 2013 or later.



[GetTexture](#)

Gets the texture applied to the specified configuration.



[GetUserPreferenceDouble](#)

Gets document default user preference values. This method is intended for user preferences of type double.



[GetUserPreferenceDoubleValueRange](#)

Gets the current document default user preference value, and the minimum and maximum valid document user preference values.



[GetUserPreferenceInteger](#)

Sets document default user preference values. This method is intended for user preferences of type integer.



[GetUserPreferenceString](#)

Gets document default user preference values. This method is intended for user preferences of type string.



[GetUserPreferenceTextFormat](#)

Gets document default user preference values. This method is intended for getting detailing text formats.



[GetUserPreferenceToggle](#)

Gets document default user preference values. This method is intended for user preferences that can be toggled.



[GetVisibleBox](#)

Gets the visible bounding box set by [IModelDocExtension::SetVisibleBox](#) for a part or an assembly.



[GetWhatsWrong](#)

Gets the What's Wrong dialog information for this model document.



[GetWhatsWrongCount](#)

Gets the number of items in the What's Wrong dialog.



[HasDesignTable](#)

Gets whether a document has a design table.



[HasLegacyCThreads](#)

Gets whether this model has legacy cosmetic threads.



[HasMaterialPropertyValues](#)

Gets whether this model has an appearance.



[HasRenamedDocuments](#)

Gets whether the document has renamed files.



[HideDecal](#)

Hides or shows the specified decal applied to this model.



[HideFeatureManager](#)

Hides or shows the Manager Pane.



[IAddDisplayStateSpecificRenderMaterial](#)

Adds the specified material to the specified display states in the active configuration and returns the IDs assigned to that material.



[IChangeSketchPlane](#)

Changes the plane used by a sketch by moving the selected sketch to the selected plane in the specified configurations.



[ICreateOLEObject](#)

Creates an OLE object on the active document.



[IDeleteDisplayStateSpecificRenderMaterial](#)

Deletes the specified materials, using the IDs of the materials, from the active configuration.



[IEditDimensionProperties](#)

Edits the selected dimension.



[IGet3rdPartyStorageStore](#)

Gets the IStorage interface to the specified third-party storage in this SOLIDWORKS document.



[IGetAnnotations](#)

Gets the annotations on this model.



[IGetAnnotationViews](#)

Gets the annotation views in this part or assembly document.



[IGetAttachments](#)

Gets the attachments for this document.



[IGetDecals](#)

Gets the decals applied to the model.



[IGetFlattenSheetMetalPersistReference](#)

Gets a byte array of persistent reference IDs for the specified entity (a face, edge, or vertex) in a flattened sheet metal part.



[IGetMassProperties](#)

Obsolete. Superseded by [IModelDocExtension::GetMassProperties2](#)



[IGetMaterialPropertyValues](#)

Gets the material properties for this model.



[IGetModelViews](#)

Gets the model views in this document.



[IGetNamedViewRotation](#)

Gets the specified named view orientation matrix with respect to the Front view.



[IGetObjectByPersistReference](#)

Obsolete. Superseded by [IModelDocExtension::IGetObjectByPersistReference3](#)

[IGetObjectByPersistReference2](#)

Obsolete. Superseded by [IModelDocExtension::IGetObjectByPersistReference3](#)

		
	IGetObjectByPersistReference3	Gets the object assigned to the specified persistent reference ID.
	IGetOLEObjects	Get the OLE objects.
	IGetPersistReference	Obsolete. Superseded by IModelDocExtension::IGetPersistReference3 .
	IGetPersistReference3	Gets the persistent reference ID for the specified object in this model document.
	IGetRenderMaterials	Obsolete. Superseded by IModelDocExtension::GetRenderMaterials2 .
	IGetSearchData	Gets the SOLIDWORKS Search, third-party, application keywords from the model document.
	IGetSectionProperties	Obsolete. Superseded by IModelDocExtension::IGetSectionProperties2 .
	IGetSectionProperties2	Gets the section properties for the following types of selected items: - Planar model face in any document - Face on a section plane - Crosshatch section face in a section view in a drawing a sketch - Sketch
	IGetSheetMetalObjectsByPersistReference	Gets the object, or objects, in a folded sheet metal part that correspond to the byte array of persistent reference IDs of an entity in a flattened sheet metal part.
	IListExternalFileReferences	Gets the names and statuses of the external file references on this part or assembly.
	IMultiSelect	Obsolete. Superseded by IModelDocExtension::MultiSelect2 .
	InsertAnnotationFavorite	Inserts annotations from the specified favorite file at the specified location.
	InsertAnnotationView	Inserts an annotation view in this part or assembly document.
	InsertAttachment	Inserts a file as an Attachment to this document.
	InsertBOMBalloon	Obsolete. Superseded by IModelDocExtension::InsertBomBalloon2 .
	InsertBOMBalloon2	Inserts a BOM balloon for the selected item.
	InsertBomTable	Obsolete. Superseded by IModelDocExtension::InsertBomTable2 .
	InsertBomTable2	Obsolete. Superseded by IModelDocExtension::InsertBomTable3 .
	InsertBomTable3	Obsolete. Superseded by IModelDocExtension::InsertBomTable4 .
	InsertBomTable4	Inserts a bill of materials (BOM) table in a part or assembly document.
	InsertCamera	Inserts a camera in this document.
	InsertChainDimensions	Chains dimensions for the specified entities in this drawing or sketch.
	InsertDatumTargetSymbol2	Obsolete. Superseded by IModelDocExtension::InsertDatumTargetSymbol3 .
	InsertDatumTargetSymbol3	Creates a datum target symbol.
	InsertDeleteFace	Inserts a DeleteFace feature.
	InsertGeneralTableAnnotation	Inserts the a general table annotation in this model document.
	InsertGeneralToleranceTableAnnotation	Inserts a general tolerance table annotation in this model document.
	InsertObjectFromFile	Inserts an OLE object from a file.
	InsertScene	Applies the specified scene to the model.
	InsertStackedBalloon	Obsolete. Superseded by IModelDocExtension::InsertStackedBalloon2 .
	InsertStackedBalloon2	Inserts a stack of balloons for selected items.
	InsertSurfaceFinishSymbol3	Creates a surface-finish symbol based on the last selection.
	InsertTitleBlockTable	Inserts a title block table in a part or assembly document.
	InstallModelColorizer	Installs your implemented interface of the ISwColorContour interface.
	IPrintOut2	Obsolete. Superseded by IModelDocExtension::IPrintOut3 .
	IPrintOut3	Obsolete. Superseded by IModelDocExtension::PrintOut4 .
	IRelease3rdPartyStorageStore	Releases the specified third-party storage in this document.
	IRemoveMaterialProperty	Removes material property values from this model.
	IsAbbreviatedViewActive	Gets or sets whether the abbreviated view is active.

	IsConverted	Gets whether the active document was converted to the current release upon opening but has not yet been saved.
	IsDisplayPaneVisible	Gets whether the Display Pane is visible in this active document.
	ISetMaterialPropertyValues	Sets the material property values for this model document.
	IsExploded	Gets the name of the exploded view currently shown in the model.
	IsFutureVersion	Gets whether this document is for a future version of SOLIDWORKS.
	IsSamePersistentID	Gets whether the two specified objects have the same persistent reference IDs.
	IsVirtualComponent	Obsolete. Superseded by IModelDocExtension::IsVirtualComponent2
	IsVirtualComponent2	Obsolete. Superseded by IModelDocExtension::IsVirtualComponent3
	IsVirtualComponent3	Gets the paths to parent assembly components, up to and including the first non-virtual parent, if the model is a virtual component.
	JogDimension	Gets or sets whether jog points are on or off on an interactively selected linear or ordinate dimension.
	ListExternalFileReferences	Obsolete. Superseded by IModelDocExtension::ListExternalFileReferences2
	ListExternalFileReferences2	Gets the specified external file reference information for this part or assembly.
	ListExternalFileReferencesCount	Gets the number of external references on this part or assembly.
	LoadDraftingStandard	Loads a custom drafting standard from a file.
	Make3DEXperienceCompatible	Makes the current model compatible with SOLIDWORKS Design connected to the 3DEXPERIENCE platform .
	MoveDecal	Moves the decal up or down in the list of decals applied to the model.
	MoveOrCopy	Moves and optionally copies the selected sketch entities or annotations.
	MultiSelect	Obsolete. Superseded by IModelDocExtension::MultiSelect2
	MultiSelect2	Selects multiple objects and returns the number of objects selected in the model.
	PrintOut	Obsolete. Superseded by IModelDocExtension::PrintOut2
	PrintOut2	Obsolete. Superseded by IModelDocExtension::PrintOut3
	PrintOut3	Obsolete. Superseded by IModelDocExtension::PrintOut4
	PrintOut4	Prints this document without displaying any dialogs or message boxes.
	PublishSTEP242File	Obsolete. Superseded by IModelDocExtension::PublishSTEP242File2
	PublishSTEP242File2	Exports the SOLIDWORKS MBD 3D part or assembly to a STEP 242 file.
	PublishTo3DPDF	Creates a 3D PDF for SOLIDWORKS MBD.
	PurgeDisplayState	Purges identical display states so that only unique display states remain.
	RayIntersections	Finds the intersections between the specified set of rays and the specified set of bodies.
	Rebuild	Rebuilds the model in assembly and drawing documents and returns the status of the rebuild.
	Refresh3DViews	Updates the 3D Views of this part or assembly.
	ReJogRunningDimension	Applies jogs where extension lines overlap dimension text in an angular running dimension.
	ReleaseStream	Releases a previously obtained stream.
	ReloadOrReplace	Reloads or replaces this model document.
	ReloadWithReferences	Reloads this model document with the specified references.
	RemoveMaterialProperty	Removes material properties from this model.
	RemoveModelColorizer	Removes your installed implemented interface of the ISwColorContour interface.
	RemoveTexture	Obsolete. Superseded by IModelDocExtension::RemoveTexture2
	RemoveTexture2	Removes the texture from the specified configuration.
	RemoveTextureByDisplayState	Removes the texture applied to this model in the specified display state.
	RemoveVisibleBox	Removes the visible bounding box set by IModelDocExtension::SetVisibleBox and resets the size of the bounding box to the size calculated by SOLIDWORKS for a part or an assembly.
	RenameDocument	Temporarily renames the selected component using the specified name.
	RenameDraftingStandard	Rename the current custom drafting to the specified name.
	ReorderFeature	Obsolete. Superseded by IModelDocExtension::ReorderFeature2

		
	ReorderFeature2	Moves the specified feature to another location in the FeatureManager design tree of this part or assembly.
	ResetStandardViews	Returns all standard model views to their default settings.
	ReverseDecalsOrder	Reverses the order of the decals on the model.
	RotateOrCopy	Rotates and optionally copies the selected sketch entities or annotations.
	RunCommand	Runs the specified SOLIDWORKS command.
	SaveAs	Obsolete. Superseded by IModelDocExtension.SaveAs2
	SaveAs2	Obsolete. Superseded by IModelDocExtension.SaveAs3
	SaveAs3	Saves the active document to the specified name with advanced options.
	SaveDeFeaturedFile	Removes all CAD data except the outer surface from a loaded part or assembly document and saves the outer surface as a part.
	SaveDraftingStandard	Saves the current custom drafting standard to a file.
	SavePackAndGo	Saves the files designated for Pack and Go to either a folder or Zip file.
	SaveSelection	Creates a new selection set containing the selected entities.
	SaveTo3DEXperience	Saves this document in SOLIDWORKS Design connected to the 3DEXPERIENCE platform using the specified save options.
	SaveViewToSolidworks	Saves the specified named view of this model to SOLIDWORKS.
	ScaleOrCopy	Scales and optionally copies the selected sketch items or annotations.
	SelectAll	Selects all edges in a part, all components in an assembly, or all entities (by default, sketch entities, dimensions, and annotations) in a drawing.
	SelectById	Obsolete. Superseded by IModelDocExtension.SelectById2
	SelectById2	Selects the specified entity.
	SelectByRay	Selects the first entity of the specified type that is intersected by a ray that starts at the specified point and runs parallel to the specified direction vector within the specified radius.
	SetAdvancedSpotLightProperties	Sets the attenuation-related, advanced properties for the specified SOLIDWORKS spot light in this model.
	SetApiUndoObject	Implements an undo object for an add-in application.
	SetKeepLightInRenderScene	Sets whether to keep a light when the scene changes.
	SetLightEnabledInRender	Sets whether a light is on in this model.
	SetMaterialPropertyValues	Sets the material property values for this model document.
	SetSceneBkgDIBx64	Sets the background image in 64-bit applications.
	SetSunLightAdvancedPropertyValues	Sets the specified sunlight advanced properties.
	SetSunLightSourcePropertyValues	Sets the property values for a sunlight source.
	SetTexture	Applies texture to the specified configuration.
	SetTextureByDisplayState	Sets the texture applied to this model in the specified display state.
	SetTopLevelTransparency	Sets the transparency of this part or top-level assembly.
	SetUserPreferenceDouble	Sets document default user preference values. This method is intended for user preferences of type double.
	SetUserPreferenceInteger	Sets document default user preference values. This method is intended for user preferences of type integer.
	SetUserPreferenceString	Sets document default user preference values. This method is intended for user preferences of type string.
	SetUserPreferenceTextFormat	Sets document default user preference values. This method is intended for setting detailing text formats.
	SetUserPreferenceToggle	Sets document default user preference values. This method is intended for user preferences that can be toggled.
	SetVisibleBox	Sets the visible bounding box for Zoom to Fit for a part or an assembly.
	ShowMessageBar	Shows the specified message bar in this document's window.
	ShowModelBreakView	Gets whether to show or hide the specified Model Break View in the current configuration of the active model.
	ShowSmartMessage	Displays a SOLIDWORKS-style message as a ToolTip in the graphics area and on the status bar.
	ShowUserNotification	Displays the specified user notification for this document.
	SketchBoxSelect	Box selects all of the entities in a sketch within the specified coordinates of the selection box.

		
	SketchOffsetOnSurface	Creates a Euclidean sketch offset from selected edges of a face or surface.
	StartFormatPainter	Starts the Format Painter.
	StartRecordingUndoObject	Starts recording the SOLIDWORKS Undo object.
	StopFormatPainter	Stops the Format Painter.
	Stretch	Stretch the selected entities.
	UpdateExternalFileReferences	Updates the external files references on this model.
	UpdateFrozenFeatures	Updates frozen features of the model.
	UpdateRenderMaterialsInSceneGraph	Sets whether to update the appearances in the graphics area in the model.
	UpdateStandardViews	Changes the specified standard view to the current model view.
	UpdateSunLight	Updates sunlight position, color, and background image.
	UpgradeLegacyCThreads	Upgrades cosmetic thread features in this legacy model to the latest cosmetic thread architecture.
	UpgradeLegacyCustomProperties	Upgrades custom properties in this legacy (created prior to SOLIDWORKS 2018) model to the latest custom properties architecture.
	ViewZoomToSheet	Zooms a drawing sheet to its maximum size within the window.

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See Also

[IModelDocExtension Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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IModelDocExtension Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IModelDocExtension Interface

For a list of all members of this type, see [IModelDocExtension members](#).

Public Methods

Name	Description
 ActivateFeatureMgrViewx64	Activates the specified FeatureManager view in x64 applications.
 AddAngularRunningDim	Adds the specified angular running dimension for selected entities.
 AddComment	Adds a comment to this document's Comment Folder.
 AddDecal	Adds a decal to the model.
 AddDefaultRenderMaterial	Not supported in SOLIDWORKS 2011 and later and not superseded.
 AddDimension	Creates a display dimension at the specified location for selected entities.
 AddDisplayStateSpecificRenderMaterial	Adds the specified appearance to the specified display states in the active configuration and returns the IDs assigned to that appearance.
 AddFeatureMgrView3x64	Adds the specified tab to the FeatureManager design tree view in x64 applications.
 AddOrdinateDimension	Inserts an ordinate dimension.
 AddOrUpdateSearchData	Adds or updates the SOLIDWORKS Search, third-party, application keyword and value to the model document.
 AddPathLengthDim	Inserts a path length dimension at the specified coordinates for a selected path.
 AddRenderMaterial	Not supported in SOLIDWORKS 2011 and later. Superseded by IModelDocExtension.AddDisplayStateSpecificRenderMaterial and IModelDocExtension.IAddDisplayStateSpecificRenderMaterial .
 AddSpecificDimension	Creates the specified display dimension at the specified location for selected entities.
 AddSymmetricDimension	Creates a full symmetrical angular dimension at the specified location for the selected entities.
 AlignDimensions	Aligns the selected dimensions in drawing documents.
 AlignRunningDimension	Aligns the extension lines of all angular dimensions to be the same distance from the center as the baseline dimension (0°) in the set of angular running dimensions.
 ApplyFormatPainterToAll	Applies Format Painter to all dimensions and annotations in the active document.

	ApplyTexture	Obsolete. Superseded by IModelDocExtension::SetTexture .
	BreakAllExternalFileReferences2	Breaks all external references and allows you to insert the features of the original part, or parts, if the external references are broken.
	Capture3DView	Captures the 3D View of this part or assembly.
	ChangeSketchPlane	Changes the plane used by a sketch by moving the selected sketch to the selected plane in the specified configurations.
	CheckVersionCompatibility	Checks whether a document can be saved in the file format for an earlier version of SOLIDWORKS.
	CloseMessageBar	Closes the specified message bar in this document's window.
	CloseUserNotification	Closes the specified user notification in this document's window.
	Compare3DPMI	Compare DimXpert annotations, reference dimensions, and other annotations between different versions of the same part document.
	CopyDraftingStandard	Copy the current custom drafting standard.
	Create3DBoundingBox	Creates a 3D bounding box for a cut list item in a weldment part.
	CreateAdvancedHoleElementData	Creates an Advanced Hole element data object of the specified type.
	CreateBalloonOptions	Creates an object that stores BOM balloon options.
	CreateCallout	Creates a callout independent of a selection.
	CreateDecal	Creates a decal for this model.
	CreateMassProperty	Obsolete. Superseded by IModelDocExtension::CreateMassProperty2 .
	CreateMassProperty2	Creates a mass properties object.
	CreateMeasure	Creates a measure tool.
	CreateOLEObject	Creates an OLE object on the active document.
	CreateRenderMaterial	Creates an appearance for this model.
	CreateStackedBalloonOptions	Creates an object that stores options for stacked balloons.
	CreateStructureSystem	Creates a structure system feature using the specified primary and secondary member arrays.
	CreateStructureSystemMemberData	Creates the specified structure system member.
	CreateTexture	Creates a texture.
	DeActivateFeatureMgrViewx64	Deactivates a tab in the FeatureManager design tree view in x64 applications.
	DeleteAllDecals	Deletes all decals on this model.
	DeleteAttachment	Deletes the specified file in the Attachments folder in the FeatureManager design tree.
	DeleteDecal	Removes the specified decal from this model.
	DeleteDisplayStateSpecificRenderMaterial	Deletes the specified appearances, using the IDs of the appearances, from the active configuration.
	DeleteDraftingStandard	Delete the current custom drafting standard.
	DeleteFeatureMgrViewx64	Removes the specified tab in the FeatureManager design tree in 64-bit applications.
	DeleteRenderMaterial	Not supported in SOLIDWORKS 2011 and later. Superseded by IModelDocExtension::DeleteDisplayStateSpecificRenderMaterial and IModelDocExtension::DeleteDisplayStateSpecificRenderMaterial2 .
	DeleteScene	Deletes the scene applied to this model.
	DeleteSearchData	Deletes the specified SOLIDWORKS Search third-party keywords from the model document.
	DeleteSelection2	Deletes the selected items, with the option to delete absorbed features, child features, or both.
	EditBalloonProperties	Obsolete. Superseded by IModelDocExtension::EditBalloonProperties2 .
	EditBalloonProperties2	Edits the selected balloon's properties.
	EditDimensionProperties	Edits the selected dimension.
	EditRebuildAll	Rebuilds only those features that need to be rebuilt in all configurations without activating each configuration.
	FindTrackedObjects	Finds the tracking IDs assigned to entities in this document.
	FinishRecordingUndoObject	Obsolete. Superseded by IModelDocExtension::FinishRecordingUndoObject2 .
	FinishRecordingUndoObject2	Ends recording of a SOLIDWORKS Undo object with the specified name and visibility.
	ForceRebuildAll	Forces a rebuild of all features in all configurations without activating each configuration.
	GeodesicSketchOffset	Creates a geodesic sketch offset along the curvature of a surface.

		
	Get3DExperienceModelType	Gets the type of 3DEXPERIENCE application that owns this model.
	Get3DView	Gets the 3D View with the specified name for this part or assembly.
	Get3DViewCount	Gets the number of 3D Views in this part or assembly.
	Get3DViewNames	Gets names of the 3D Views in this part or assembly.
	Get3DViews	Gets the 3D Views for this part or assembly.
	GetActiveAnnotationView	Gets the active annotation view of this model.
	GetActivePropertyManagerPage	Gets the name of the active PropertyManager page.
	GetAdvancedSaveAsOptions	Gets the advanced File > Save As options.
	GetAdvancedSpotLightProperties	Gets the attenuation-related, advanced properties for the specified SOLIDWORKS spot light in this model.
	GetAnnotationCount	Gets the number of annotations on this part.
	GetAnnotations	Gets the annotations on this part.
	GetAnnotationsByType	Gets the annotations of the specified type in this document.
	GetAppearanceSetting	Gets the appearance setting for this document.
	GetAttachmentCount	Gets the number of attachments for this document.
	GetAttachments	Gets the attachments for this document.
	GetCalloutVariableString	Gets the string for the specified callout variable.
	GetCameraById	Gets a camera using the specified camera ID.
	GetCameraCount	Gets the number of cameras in the document.
	GetCameraDefinition	Gets a camera, but does not add the newly created camera to the model.
	GetCommandTabs	Gets all of the SOLIDWORKS CommandManager tab names in this document.
	GetCoordinateSystemTransformByName	Gets the transform of the specified coordinate system.
	GetCorresponding	Obsolete. Superseded by IModelDocExtension.GetCorresponding2 .
	GetCorresponding2	Gets the object in the underlying part or subassembly document that corresponds to the specified input object in this drawing view or assembly.
	GetCorrespondingEntity	Obsolete. Superseded by IModelDocExtension.GetCorrespondingEntity2 .
	GetCorrespondingEntity2	Gets the entity in the underlying part or subassembly that corresponds to the specified entity in this assembly or drawing view.
	GetCostingManager	Gets the entry-point interface to the SOLIDWORKS Costing API and gets the CostingManager.
	GetCSYSDistances	Gets the specified distances between two selected coordinate systems.
	GetCSYSEulersAngularRotation	Gets the specified Eulers angular rotation values that transform one selected coordinate system into another.
	GetCSYSXYZAngularRotation	Gets the specified Tait-Bryan angular rotation values that transform one selected coordinate system into another.
	GetDecal	Gets the specified decal in this model.
	GetDecals	Gets the decals applied to the model.
	GetDecalsCount	Gets the number of decals applied to this model.
	GetDependencies	Gets all of this model's dependencies.
	GetDisplayStateSetting	Gets the display state setting for the specified scope.
	GetDraftingStandardNames	Get the names of all currently available drafting standards.
	GetFlattenSheetMetalPersistReference	Gets a byte array of persistent reference IDs for the specified entity (a face, edge, or vertex) in a flattened sheet metal part.
	GetGeneralTableAnnotation	Creates a general table annotation for SOLIDWORKS MBD 3D PDF.
	GetKeepLightInRenderScene	Gets whether a light is kept when the scene changes.
	GetLastFeatureAdded	Gets the last feature added to the model.
	GetLicenseType	Gets the type of SOLIDWORKS license used when the model was created.
	GetLightEnabledInRender	Gets whether a light is on in this model.
	GetMassProperties	Obsolete. Superseded by IModelDocExtension.GetMassProperties2 .

		
	GetMassProperties2	Obsolete. Superseded by IMassProperty2
	GetMaterial	Gets the appearance for the specified appearance ID in the specified configuration in this model document
	GetMaterialPropertyValues	Gets the material properties for this model document.
	GetMBD3DPdfData	Gets the SOLIDWORKS MBD 3D PDF data object.
	GetMBDSTEP242Data	Gets the SOLIDWORKS MBD STEP 242 data object.
	GetModelBreakViewNames	Gets the names and number of the Model Break Views in the current configuration of the active model.
	GetModelViewCount	Gets the number of model views in this document.
	GetModelViews	Gets the model views in this document.
	GetMotionStudyManager	Gets the SOLIDWORKS motion study's MotionManager.
	GetNamedViewRotation	Gets the specified named view orientation matrix with respect to the Front view.
	GetObjectByPersistReference	Obsolete. Superseded by IModelDocExtension::GetObjectByPersistReference3
	GetObjectByPersistReference2	Obsolete. Superseded by IModelDocExtension::GetObjectByPersistReference3
	GetObjectByPersistReference3	Gets the object assigned to the specified persistent reference ID.
	GetObjectId	Gets the object ID for the specified annotation.
	GetOLEObjectCount	Gets the number of OLE objects.
	GetOLEObjects	Get the OLE objects.
	GetPackAndGo	Gets a Pack and Go object.
	GetPersistReference	Obsolete. Superseded by IModelDocExtension::GetPersistReference3
	GetPersistReference3	Gets the persistent reference ID for the specified object in this model document.
	GetPersistReferenceCount	Obsolete. Superseded by IModelDocExtension::GetPersistReferenceCount3
	GetPersistReferenceCount3	Gets the size of the persistent reference ID assigned to the selected object in this model document.
	GetPLMID	Gets the ID of this SOLIDWORKS Design connected to the 3DEXPERIENCE platform document.
	GetPrint3DDialog	Gets the IPrint3DDialog object.
	GetPrintSpecification	Gets the IPrintSpecification object for this document.
	GetRenderCustomReferences	Get the custom render references for this model.
	GetRenderMaterials	Obsolete. Superseded by IModelDocExtension::GetRenderMaterials2
	GetRenderMaterials2	Gets the appearances applied to this model document in the specified display states.
	GetRenderMaterialsCount	Obsolete. Superseded by IModelDocExtension::GetRenderMaterialsCount2
	GetRenderMaterialsCount2	Gets the number of appearances applied to this model document in the specified display states.
	GetRenderStockReferences	Gets the SOLIDWORKS-supplied (stock) render references for this model.
	GetRoutingComponentManager	Gets the routing component manager.
	GetScanto3D	Gets the IScanTo3D object for this document.
	GetSceneBkgDIBx64	Gets the background image as DIBSECTION in 64-bit applications.
	GetSearchData	Gets the SOLIDWORKS Search, third-party, application keywords from the model document.
	GetSearchDataCount	Gets the number of SOLIDWORKS Search keywords for the specified third-party application previously added to this model document.
	GetSectionProperties	Obsolete. Superseded by IModelDocExtension::GetSectionProperties2
	GetSectionProperties2	Gets the section properties for the following types of selected items: - Planar model face in any document - Face on a section plane - Crosshatch section face in a section view in a drawing a sketch - Sketch
	GetSheetMetalObjectsByPersistReference	Gets the objects in a folded sheet metal part that correspond to the byte array of persistent reference IDs of an entity in a flattened sheet metal part.
	GetSphericalBoxDiameter	Gets the diameter of the spherical bounding box of this model.

	GetStream	Gets the handle for the specified stream.
	GetSunLightAdvancedPropertyValues	Gets the specified sunlight advanced properties.
	GetSunLightSourcePropertyValues	Gets the property values for a sunlight source.
	GetSustainability	Gets the entry-point interface to the SOLIDWORKS Sustainability API.
	GetTemplateSheetMetal	Gets the sheet metal folder feature from this sheet metal model created in SOLIDWORKS 2013 or later.
	GetTexture	Gets the texture applied to the specified configuration.
	GetUserPreferenceDouble	Gets document default user preference values. This method is intended for user preferences of type double.
	GetUserPreferenceDoubleValueRange	Gets the current document default user preference value, and the minimum and maximum valid document user preference values.
	GetUserPreferenceInteger	Sets document default user preference values. This method is intended for user preferences of type integer.
	GetUserPreferenceString	Gets document default user preference values. This method is intended for user preferences of type string.
	GetUserPreferenceTextFormat	Gets document default user preference values. This method is intended for getting detailing text formats.
	GetUserPreferenceToggle	Gets document default user preference values. This method is intended for user preferences that can be toggled.
	GetVisibleBox	Gets the visible bounding box set by IModelDocExtension::SetVisibleBox for a part or an assembly.
	GetWhatsWrong	Gets the What's Wrong dialog information for this model document.
	GetWhatsWrongCount	Gets the number of items in the What's Wrong dialog.
	HasDesignTable	Gets whether a document has a design table.
	HasLegacyCThreads	Gets whether this model has legacy cosmetic threads.
	HasMaterialPropertyValues	Gets whether this model has an appearance.
	HasRenamedDocuments	Gets whether the document has renamed files.
	HideDecal	Hides or shows the specified decal applied to this model.
	HideFeatureManager	Hides or shows the Manager Pane.
	IAddDisplayStateSpecificRenderMaterial	Adds the specified material to the specified display states in the active configuration and returns the IDs assigned to that material.
	IChangeSketchPlane	Changes the plane used by a sketch by moving the selected sketch to the selected plane in the specified configurations.
	ICreateOLEObject	Creates an OLE object on the active document.
	IDeleteDisplayStateSpecificRenderMaterial	Deletes the specified materials, using the IDs of the materials, from the active configuration.
	IEditDimensionProperties	Edits the selected dimension.
	IGet3rdPartyStorageStore	Gets the IStorage interface to the specified third-party storage in this SOLIDWORKS document.
	IGetAnnotations	Gets the annotations on this model.
	IGetAnnotationViews	Gets the annotation views in this part or assembly document.
	IGetAttachments	Gets the attachments for this document.
	IGetDecals	Gets the decals applied to the model.
	IGetFlattenSheetMetalPersistReference	Gets a byte array of persistent reference IDs for the specified entity (a face, edge, or vertex) in a flattened sheet metal part.
	IGetMassProperties	Obsolete. Superseded by IModelDocExtension::GetMassProperties2 .
	IGetMaterialPropertyValues	Gets the material properties for this model.
	IGetModelViews	Gets the model views in this document.
	IGetNamedViewRotation	Gets the specified named view orientation matrix with respect to the Front view.
	IGetObjectByPersistReference	Obsolete. Superseded by IModelDocExtension::IGetObjectByPersistReference3 .
	IGetObjectByPersistReference2	Obsolete. Superseded by IModelDocExtension::IGetObjectByPersistReference3 .
	IGetObjectByPersistReference3	Gets the object assigned to the specified persistent reference ID.
	IGetOLEObjects	Get the OLE objects.
	IGetPersistReference	Obsolete. Superseded by IModelDocExtension::IGetPersistReference3 .
	IGetPersistReference3	Gets the persistent reference ID for the specified object in this model document.
	IGetRenderMaterials	Obsolete. Superseded by IModelDocExtension::GetRenderMaterials2 .

		
	IGetSearchData	Gets the SOLIDWORKS Search, third-party, application keywords from the model document.
	IGetSectionProperties	Obsolete. Superseded by IModelDocExtension::IGetSectionProperties2 .
	IGetSectionProperties2	Gets the section properties for the following types of selected items: - Planar model face in any document - Face on a section plane - Crosshatch section face in a section view in a drawing a sketch - Sketch
	IGetSheetMetalObjectsByPersistReference	Gets the object, or objects, in a folded sheet metal part that correspond to the byte array of persistent reference IDs of an entity in a flattened sheet metal part.
	IListExternalFileReferences	Gets the names and statuses of the external file references on this part or assembly.
	IMultiSelect	Obsolete. Superseded by IModelDocExtension::IMultiSelect2 .
	InsertAnnotationFavorite	Inserts annotations from the specified favorite file at the specified location.
	InsertAnnotationView	Inserts an annotation view in this part or assembly document.
	InsertAttachment	Inserts a file as an Attachment to this document.
	InsertBOMBalloon	Obsolete. Superseded by IModelDocExtension::InsertBomBalloon2 .
	InsertBOMBalloon2	Inserts a BOM balloon for the selected item.
	InsertBomTable	Obsolete. Superseded by IModelDocExtension::InsertBomTable2 .
	InsertBomTable2	Obsolete. Superseded by IModelDocExtension::InsertBomTable3 .
	InsertBomTable3	Obsolete. Superseded by IModelDocExtension::InsertBomTable4 .
	InsertBomTable4	Inserts a bill of materials (BOM) table in a part or assembly document.
	InsertCamera	Inserts a camera in this document.
	InsertChainDimensions	Chains dimensions for the specified entities in this drawing or sketch.
	InsertDatumTargetSymbol2	Obsolete. Superseded by IModelDocExtension::InsertDatumTargetSymbol3 .
	InsertDatumTargetSymbol3	Creates a datum target symbol.
	InsertDeleteFace	Inserts a DeleteFace feature.
	InsertGeneralTableAnnotation	Inserts the a general table annotation in this model document.
	InsertGeneralToleranceTableAnnotation	Inserts a general tolerance table annotation in this model document.
	InsertObjectFromFile	Inserts an OLE object from a file.
	InsertScene	Applies the specified scene to the model.
	InsertStackedBalloon	Obsolete. Superseded by IModelDocExtension::InsertStackedBalloon2 .
	InsertStackedBalloon2	Inserts a stack of balloons for selected items.
	InsertSurfaceFinishSymbol3	Creates a surface-finish symbol based on the last selection.
	InsertTitleBlockTable	Inserts a title block table in a part or assembly document.
	InstallModelColorizer	Installs your implemented interface of the ISwColorContour interface.
	IPrintOut2	Obsolete. Superseded by IModelDocExtension::IPrintOut3 .
	IPrintOut3	Obsolete. Superseded by IModelDocExtension::PrintOut4 .
	IRelease3rdPartyStorageStore	Releases the specified third-party storage in this document.
	IRemoveMaterialProperty	Removes material property values from this model.
	IsAbbreviatedViewActive	Gets or sets whether the abbreviated view is active.
	IsConverted	Gets whether the active document was converted to the current release upon opening but has not yet been saved.
	IsDisplayPaneVisible	Gets whether the Display Pane is visible in this active document.
	ISetMaterialPropertyValues	Sets the material property values for this model document.
	IsExploded	Gets the name of the exploded view currently shown in the model.
	IsFutureVersion	Gets whether this document is for a future version of SOLIDWORKS.

	IsSamePersistentID	Gets whether the two specified objects have the same persistent reference IDs.
	IsVirtualComponent	Obsolete. Superseded by IModelDocExtension::IsVirtualComponent2 .
	IsVirtualComponent2	Obsolete. Superseded by IModelDocExtension::IsVirtualComponent3 .
	IsVirtualComponent3	Gets the paths to parent assembly components, up to and including the first non-virtual parent, if the model is a virtual component.
	JogDimension	Gets or sets whether jog points are on or off on an interactively selected linear or ordinate dimension.
	ListExternalFileReferences	Obsolete. Superseded by IModelDocExtension::ListExternalFileReferences2 .
	ListExternalFileReferences2	Gets the specified external file reference information for this part or assembly.
	ListExternalFileReferencesCount	Gets the number of external references on this part or assembly.
	LoadDraftingStandard	Loads a custom drafting standard from a file.
	Make3DEXperienceCompatible	Makes the current model compatible with SOLIDWORKS Design connected to the 3DEXPERIENCE platform .
	MoveDecal	Moves the decal up or down in the list of decals applied to the model.
	MoveOrCopy	Moves and optionally copies the selected sketch entities or annotations.
	MultiSelect	Obsolete. Superseded by IModelDocExtension::MultiSelect2 .
	MultiSelect2	Selects multiple objects and returns the number of objects selected in the model.
	PrintOut	Obsolete. Superseded by IModelDocExtension::PrintOut2 .
	PrintOut2	Obsolete. Superseded by IModelDocExtension::PrintOut3 .
	PrintOut3	Obsolete. Superseded by IModelDocExtension::PrintOut4 .
	PrintOut4	Prints this document without displaying any dialogs or message boxes.
	PublishSTEP242File	Obsolete. Superseded by IModelDocExtension::PublishSTEP242File2 .
	PublishSTEP242File2	Exports the SOLIDWORKS MBD 3D part or assembly to a STEP 242 file.
	PublishTo3DPDF	Creates a 3D PDF for SOLIDWORKS MBD.
	PurgeDisplayState	Purges identical display states so that only unique display states remain.
	RayIntersections	Finds the intersections between the specified set of rays and the specified set of bodies.
	Rebuild	Rebuilds the model in assembly and drawing documents and returns the status of the rebuild.
	Refresh3DViews	Updates the 3D Views of this part or assembly.
	ReJogRunningDimension	Applies jogs where extension lines overlap dimension text in an angular running dimension.
	ReleaseStream	Releases a previously obtained stream.
	ReloadOrReplace	Reloads or replaces this model document.
	ReloadWithReferences	Reloads this model document with the specified references.
	RemoveMaterialProperty	Removes material properties from this model.
	RemoveModelColorizer	Removes your installed implemented interface of the ISwColorContour interface.
	RemoveTexture	Obsolete. Superseded by IModelDocExtension::RemoveTexture2 .
	RemoveTexture2	Removes the texture from the specified configuration.
	RemoveTextureByDisplayState	Removes the texture applied to this model in the specified display state.
	RemoveVisibleBox	Removes the visible bounding box set by IModelDocExtension::SetVisibleBox and resets the size of the bounding box to the size calculated by SOLIDWORKS for a part or an assembly.
	RenameDocument	Temporarily renames the selected component using the specified name.
	RenameDraftingStandard	Rename the current custom drafting to the specified name.
	ReorderFeature	Obsolete. Superseded by IModelDocExtension::ReorderFeature2 .
	ReorderFeature2	Moves the specified feature to another location in the FeatureManager design tree of this part or assembly.
	ResetStandardViews	Returns all standard model views to their default settings.
	ReverseDecalsOrder	Reverses the order of the decals on the model.
	RotateOrCopy	Rotates and optionally copies the selected sketch entities or annotations.
	RunCommand	Runs the specified SOLIDWORKS command.

		
	SaveAs	Obsolete. Superseded by IModelDocExtension.SaveAs2
	SaveAs2	Obsolete. Superseded by IModelDocExtension.SaveAs3
	SaveAs3	Saves the active document to the specified name with advanced options.
	SaveDeFeaturedFile	Removes all CAD data except the outer surface from a loaded part or assembly document and saves the outer surface as a part.
	SaveDraftingStandard	Saves the current custom drafting standard to a file.
	SavePackAndGo	Saves the files designated for Pack and Go to either a folder or Zip file.
	SaveSelection	Creates a new selection set containing the selected entities.
	SaveTo3DEXperience	Saves this document in SOLIDWORKS Design connected to the 3DEXPERIENCE platform using the specified save options.
	SaveViewToSolidworks	Saves the specified named view of this model to SOLIDWORKS.
	ScaleOrCopy	Scales and optionally copies the selected sketch items or annotations.
	SelectAll	Selects all edges in a part, all components in an assembly, or all entities (by default, sketch entities, dimensions, and annotations) in a drawing.
	SelectById	Obsolete. Superseded by IModelDocExtension.SelectById2
	SelectById2	Selects the specified entity.
	SelectByRay	Selects the first entity of the specified type that is intersected by a ray that starts at the specified point and runs parallel to the specified direction vector within the specified radius.
	SetAdvancedSpotLightProperties	Sets the attenuation-related, advanced properties for the specified SOLIDWORKS spot light in this model.
	SetApiUndoObject	Implements an undo object for an add-in application.
	SetKeepLightInRenderScene	Sets whether to keep a light when the scene changes.
	SetLightEnabledInRender	Sets whether a light is on in this model.
	SetMaterialPropertyValues	Sets the material property values for this model document.
	SetSceneBgDIBv64	Sets the background image in 64-bit applications.
	SetSunLightAdvancedPropertyValues	Sets the specified sunlight advanced properties.
	SetSunLightSourcePropertyValues	Sets the property values for a sunlight source.
	SetTexture	Applies texture to the specified configuration.
	SetTextureByDisplayState	Sets the texture applied to this model in the specified display state.
	SetTopLevelTransparency	Sets the transparency of this part or top-level assembly.
	SetUserPreferenceDouble	Sets document default user preference values. This method is intended for user preferences of type double.
	SetUserPreferenceInteger	Sets document default user preference values. This method is intended for user preferences of type integer.
	SetUserPreferenceString	Sets document default user preference values. This method is intended for user preferences of type string.
	SetUserPreferenceTextFormat	Sets document default user preference values. This method is intended for setting detailing text formats.
	SetUserPreferenceToggle	Sets document default user preference values. This method is intended for user preferences that can be toggled.
	SetVisibleBox	Sets the visible bounding box for Zoom to Fit for a part or an assembly.
	ShowMessageBar	Shows the specified message bar in this document's window.
	ShowModelBreakView	Gets whether to show or hide the specified Model Break View in the current configuration of the active model.
	ShowSmartMessage	Displays a SOLIDWORKS-style message as a ToolTip in the graphics area and on the status bar.
	ShowUserNotification	Displays the specified user notification for this document.
	SketchBoxSelect	Box selects all of the entities in a sketch within the specified coordinates of the selection box.
	SketchOffsetOnSurface	Creates a Euclidean sketch offset from selected edges of a face or surface.
	StartFormatPainter	Starts the Format Painter.
	StartRecordingUndoObject	Starts recording the SOLIDWORKS Undo object.
	StopFormatPainter	Stops the Format Painter.
	Stretch	Stretch the selected entities.

		
	UpdateExternalFileReferences	Updates the external files references on this model.
	UpdateFrozenFeatures	Updates frozen features of the model.
	UpdateRenderMaterialsInSceneGraph	Sets whether to update the appearances in the graphics area in the model.
	UpdateStandardViews	Changes the specified standard view to the current model view.
	UpdateSunLight	Updates sunlight position, color, and background image.
	UpgradeLegacyCThreads	Upgrades cosmetic thread features in this legacy model to the latest cosmetic thread architecture.
	UpgradeLegacyCustomProperties	Upgrades custom properties in this legacy (created prior to SOLIDWORKS 2018) model to the latest custom properties architecture.
	ViewZoomToSheet	Zooms a drawing sheet to its maximum size within the window.

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See Also

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[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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IModelDocExtension Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IModelDocExtension Interface

For a list of all members of this type, see [IModelDocExtension members](#).

Public Properties

Name	Description
 ActiveCommandTab	Gets and sets the active SOLIDWORKS CommandManager tab.
 ActiveCommandTabIndex	Gets and sets the index of the active SOLIDWORKS CommandManager tab.
 AnnotationViewCount	Gets the number of annotation views in this part or assembly document.
 AnnotationViews	Gets the annotation views in this part or assembly document.
 AppPageSetup	Gets the SOLIDWORKS Design application page setup interface for this document.
 CommandTabVisible	Gets and sets the visibility of the specified SOLIDWORKS CommandManager tab.
 CustomPropertyBuilderTemplate	Gets or sets the custom property builder template for this part.
 CustomPropertyManager	Gets the custom properties for this document or configuration.
 DimXpertManager	Gets the DimXpert schema for this configuration.
 DisplayMode	Gets and sets the display modes for the specified display state setting.
 DisplayStateSpecMaterialPropertyValues	Gets and sets the appearance settings for the specified display state setting.
 DisplayTitle	Gets and sets this model's title to display in the FeatureManager design tree.
 Document	Gets the model document.
 FeatureManagerFilterString	Gets or sets the string in the FeatureManager design tree filter.
 FlyoutFeatureTreeVisibility	Gets or sets the state of the flyout FeatureManager design tree.
 HasLegacyCustomProperties	Gets whether this model has legacy custom properties.
 IncludeMassPropertiesOfHiddenBodies	Gets or sets whether to include the mass properties of hidden components in the assembly.
 LinkedDisplayState	Gets or sets whether a display state is linked in this part.
 NeedsRebuild	Obsolete. Superseded by IModelDocExtension.NeedsRebuild2.
 NeedsRebuild2	Gets whether the model document needs to be rebuilt.
 ShowPartRebuildIndicators	Gets or sets whether to display rebuild indicators on parts that have out-of-date frozen features.
 SunLightInformation	Gets the specified sunlight information.

	ToolboxPartType	Gets and sets whether this part is a SOLIDWORKS Toolbox part.
	Transparency	Gets and sets the transparency states for the specified display state setting.
	UsePageSetup	Gets or sets whether this document uses its own page setup values, SOLIDWORKS Design application page setup values, or setup values on individual drawing sheets.
	ViewDisplayRealView	Gets or sets the RealView Graphics setting.
	Visibility	Gets and sets the visibility states for the specified display state setting.

[Top](#)

See Also

[IModelDocExtension Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ActivateFeatureMgrViewx64 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ActivateFeatureMgrViewx64 Method (IModelDocExtension)

Activates the specified FeatureManager view in x64 applications.

.NET Syntax

Visual Basic (Declaration)

```
Function ActivateFeatureMgrViewx64( _
    ByRef AppView As System.Long _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim AppView As System.Long
Dim value As System.Integer

value = instance.ActivateFeatureMgrViewx64(AppView)
```

C#

```
System.int ActivateFeatureMgrViewx64(
    ref System.long AppView
)
```

C++/CLI

```
System.int ActivateFeatureMgrViewx64(
    & System.int64% AppView
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AppView

Pointer to a FeatureManager design tree view

Return Value

Pane as defined by [swFeatMgrPane_e](#) where the view resides

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ActivateFeatureMgrViewx64](#).

Remarks

See [IFeatMgrView::ActivateView](#) for examples you can tailor to use this method in x64 applications.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2026 SP01, Revision Number 34.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

ActiveCommandTab Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ActiveCommandTab Property (IModelDocExtension)

Gets and sets the active SOLIDWORKS CommandManager tab.

.NET Syntax

Visual Basic (Declaration)

```
Property ActiveCommandTab As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.String
```

```
instance.ActiveCommandTab = value
```

```
value = instance.ActiveCommandTab
```

C#

```
System.string ActiveCommandTab {get; set;}
```

C++/CLI

```
property System.String^ ActiveCommandTab {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Active tab name

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ActiveCommandTab](#).

Example

[Activate SOLIDWORKS CommandManager Tab \(VBA\)](#)
[Activate SOLIDWORKS CommandManager Tab \(VB.NET\)](#)
[Activate SOLIDWORKS CommandManager Tab \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetCommandTabs Method \(\)](#)
[IModelDocExtension::ActiveCommandTabIndex Property \(\)](#)
[IModelDocExtension::CommandTabVisible Property \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

ActiveCommandTabIndex Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ActiveCommandTabIndex Property (IModelDocExtension)

Gets and sets the index of the active SOLIDWORKS CommandManager tab.

.NET Syntax

Visual Basic (Declaration)

Property ActiveCommandTabIndex As System.Integer

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim value As System.Integer

instance.ActiveCommandTabIndex = value

value = instance.ActiveCommandTabIndex
```

C#

```
System.int ActiveCommandTabIndex {get; set;}
```

C++/CLI

```
property System.int ActiveCommandTabIndex {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Index of the active tab

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ActiveCommandTabIndex](#).

Example

[Activate SOLIDWORKS CommandManager Tab \(VBA\)](#)
[Activate SOLIDWORKS CommandManager Tab \(VB.NET\)](#)
[Activate SOLIDWORKS CommandManager Tab \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::ActiveCommandTab Property \(\)](#)
[IModelDocExtension::CommandTabVisible Property \(\)](#)
[IModelDocExtension::GetCommandTabs Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

AddAngularRunningDim Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddAngularRunningDim Method (IModelDocExtension)

Adds the specified angular running dimension for selected entities.

.NET Syntax

```

Function AddAngularRunningDim( _
    ByVal DisplayAsChain As System.Boolean, _
    ByVal RunBidirectionally As System.Boolean, _
    ByVal ExtensionLineExtendsFromCenterOfSet As System.Boolean, _
    ByVal LocX As System.Double, _
    ByVal LocY As System.Double, _
    ByVal LocZ As System.Double, _
    ByRef RetVal As System.Integer _
) As System.Object

```



```

Dim instance As IModelDocExtension
Dim DisplayAsChain As System.Boolean
Dim RunBidirectionally As System.Boolean
Dim ExtensionLineExtendsFromCenterOfSet As System.Boolean
Dim LocX As System.Double
Dim LocY As System.Double
Dim LocZ As System.Double
Dim Retval As System.Integer
Dim value As System.Object

value = instance.AddAngularRunningDim(DisplayAsChain, RunBidirectionally,
ExtensionLineExtendsFromCenterOfSet, LocX, LocY, LocZ, Retval)

```

```

System.object AddAngularRunningDim(
    System.bool DisplayAsChain,
    System.bool RunBidirectionally,
    System.bool ExtensionLineExtendsFromCenterOfSet,
    System.double LocX,
    System.double LocY,
    System.double LocZ,
    out System.int RetVal
)

```

```

System.Object^ AddAngularRunningDim(
    & System.bool DisplayAsChain,
    & System.bool RunBidirectionally,
    & System.bool ExtensionLineExtendsFromCenterOfSet,
    & System.double LocX,
    & System.double LocY,
    & System.double LocZ,
    & [Out] System.int RetVal
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DisplayAsChain

True to chain the angular dimensions together, false to not

RunBidirectionally

True if each angular dimension runs in a direction closest to the baseline, false if all angular dimensions run in the same direction (see **Remarks**)

ExtensionLineExtendsFromCenterOfSet

True to extend the extension lines from the center of the set of angular running dimensions, false to not

LocX

X coordinate of baseline dimension (see **Remarks**)

LocY

Y coordinate of baseline dimension (see **Remarks**)

LocZ

Z coordinate of baseline dimension (see **Remarks**)

RetVal

Error as defined by [swCreateAngRunDimError_e](#)

Return Value

Array of [IDisplayDimensions](#)

See [ModelDocExtension::AddAngularRunningDim](#).

Example

[Insert Angular Running Dimension \(VBA\)](#)
[Insert Angular Running Dimension \(VB.NET\)](#)
[Insert Angular Running Dimension \(C#\)](#)

Remarks

Before calling this method, call [IModelDocExtension::SelectByID2](#) with Append set to true to select the baseline entity and all entities whose angles from the baseline entity you want to measure and display.

A vertical line is constructed between [LocX, LocY, LocZ] and the baseline entity. The baseline dimension (0°) is displayed at [LocX, LocY, LocZ].

If RunBidirectionally is...	Then...
True	No angle dimension in the angular running dimension is greater than 180° from the baseline, and each angle is measured from a direction that is closest to the baseline.
False	The running direction of all angle dimensions is determined by the first angle dimension added. For example, if the user places the baseline dimension at the top of the part, clicking on a feature to the right of the baseline dimension forces all subsequent angular dimensions to be measured clockwise from the baseline.

Selections made after this method is called add angular dimensions to the set. When you have finished adding angular dimensions to the set, call [IModelDoc2::SetPickMode](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IDrawingDoc::InsertAngularRunningDim Method \(\)](#)
[IModelDocExtension::AlignRunningDimension Method \(\)](#)
[IModelDocExtension::ReJogRunningDimension Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddComment Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddComment Method (IModelDocExtension)

Adds a comment to this document's Comment Folder.

.NET Syntax

Visual Basic (Declaration)

```
Function AddComment( _  
    ByVal Text As System.String _  
) As Comment
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim Text As System.String  
Dim value As Comment  
  
value = instance.AddComment(Text)
```

C#

```
Comment AddComment(  
    System.string Text  
)
```

C++/CLI

```
Comment^ AddComment(  
& System.String^ Text  
)
```



NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Text

Comment to add to the document's Comment folder

Return Value

[Comment](#) folder

 Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddComment](#).

 Example

[Add Comment to Assembly Component \(VBA\)](#)
[Add Comment to Assembly Component \(VB.NET\)](#)
[Add Comment to Assembly Component \(C#\)](#)

 Remarks

A comment is added to the document's Comment folder, and the comment is associated with the preselected item when this method is run.
For example:

If...	Then the comment is...
A feature is preselected	Associated with the feature (equivalent to running the IFeature::AddComment on the selected feature)
The Comment folder is preselected	Not associated with any feature, but is added to the Comment folder (equivalent to running the ICommentFolder::AddComment method)

Although it might appear that ModelDocExtension::AddComment duplicates IFeature::AddComment and ICommentFolder::AddComment, IModelDocExtension::AddComment is used when recording macros, so it is different. IModelDocExtension::AddComment shares code with the user interface, so it should behave identical to the user interface.

IModelDocExtension::AddComment rebuilds the FeatureManager design tree automatically, which can be a time-consuming operation if the FeatureManager design tree gets large. Take this into consideration before using it.

 See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IComment::Delete Method](#)

 Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddDecal Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddDecal Method (IModelDocExtension)

Adds a decal to the model.

 .NET Syntax

Visual Basic (Declaration)

```
Function AddDecal(  
    ByVal pDecal As Decal, _  
    ByRef DecalID As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim PDecal As Decal
Dim DecalID As System.Integer
Dim value As System.Boolean

value = instance.AddDecal(PDecal, DecalID)
```

C#

```
System.bool AddDecal(
    Decal PDecal,
    out System.int DecalID
)
```

C++/CLI

```
System.bool AddDecal(
    & Decal^ PDecal,
    & [Out] System.int DecalID
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PDecal
[Decal](#)
DecalID
Decal ID

Return Value

True if the decal was added to the model, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddDecal](#).

Example

[Add Decal \(VBA\)](#)
[Add Decal \(C#\)](#)
[Add Decal \(VB.NET\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IDecal Interface](#)
[IModelDocExtension::CreateDecal Method](#)
[IModelDocExtension::DeleteAllDecals Method](#)
[IModelDocExtension::DeleteDecal Method](#)
[IModelDocExtension::GetDecal Method](#)
[IModelDocExtension::GetDecals Method](#)
[IModelDocExtension::GetDecalsCount Method](#)
[IModelDocExtension::HideDecal Method](#)
[IModelDocExtension::GetDecals Method](#)
[IModelDocExtension::MoveDecal Method](#)
[IModelDocExtension::ReverseDecalsOrder Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddDefaultRenderMaterial Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddDefaultRenderMaterial Method (IModelDocExtension)

Not supported in SOLIDWORKS 2011 and later and not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Function AddDefaultRenderMaterial( _
    ByVal PRenderMaterial As RenderMaterial, _
    ByRef PwMaterialId As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim PRenderMaterial As RenderMaterial
Dim PwMaterialId As System.Integer
Dim value As System.Boolean

value = instance.AddDefaultRenderMaterial(PRenderMaterial, PwMaterialId)
```

C#

```
System.bool AddDefaultRenderMaterial(
    RenderMaterial PRenderMaterial,
    out System.int PwMaterialId
)
```

C++/CLI

```
System.bool AddDefaultRenderMaterial(
    & RenderMaterial^ PRenderMaterial,
    [Out] System.int PwMaterialId
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PRenderMaterial
[Appearance](#) to add to the model
PwMaterialId
 Appearance ID

Return Value

True if the appearance is added to the model document, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddDefaultRenderMaterial](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddDimension Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddDimension Method (IModelDocExtension)

Creates a display dimension at the specified location for selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Function AddDimension( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double, _
```

```
ByVal Direction As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim Direction As System.Integer  
Dim value As System.Object  
  
value = instance.AddDimension(X, Y, Z, Direction)
```

C#

```
System.object AddDimension(  
    System.double X,  
    System.double Y,  
    System.double Z,  
    System.int Direction  
)
```

C++/CLI

```
System.Object^ AddDimension(  
& System.double X,  
& System.double Y,  
& System.double Z,  
& System.int Direction  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X

X coordinate of display dimension text

Y

Y coordinate of display dimension text

Z

Z coordinate of display dimension text

Direction

Direction of dimensioning extension line or rapid dimensioning quadrant as defined in [swSmartDimensionDirection_e](#) (see **Remarks**)

Return Value

[IDisplayDimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddDimension](#).

Example

[Create Angular Dimension \(VBA\)](#)
[Create Angular Dimension \(VB.NET\)](#)
[Create Angular Dimension \(C#\)](#)

Remarks

The X, Y, and Z coordinates of the display dimension text must be appropriate for the specified Direction, or this method fails to add the display dimension.

For parts, Direction specifies the dimensioning manipulator direction that appears in the user interface when an extension line is needed to unambiguously define what to dimension.

For drawings, Direction specifies the rapid dimensioning selector quadrant to which to add the display dimension.

Before calling this method, you must select the entities for which to add a display dimension. Entities must be selected by location and not name. If you specify line names instead of coordinates, the dimensioning routines randomly pick locations on the lines, which can yield unpredictable results during angular dimension creation.

For example, to create an angular dimension between two lines:

1. Call [IModelDocExtension::SelectByID2](#) to select a sketch segment of the angle you want to dimension, specifying the line's X, Y, and Z coordinates.
2. Call [IModelDocExtension::SelectByID2](#) to select the vertex of the angle you want to dimension, specifying the vertex's X, Y, and Z coordinates.
3. Call this method, specifying the X, Y, Z coordinates of the display dimension text and the Direction of the extension line that is needed to unambiguously define the angle to dimension.

If the pre-selected entities unambiguously define what you wish to dimension, then no extension lines are needed. In that case, call [IModelDoc2::AddDimension2](#) instead of this method.

NOTE: To change the angle in an angular dimension to its supplementary angle, use [IDisplayDimension::SupplementaryAngle](#).

You should only use this method on visible documents. Before using this method, use [ISldWorks::Visible](#) to check whether a document is visible

Before calling this method, you might want to call [ISldWorks::SetUserPreferenceToggle](#) with [swUserPreferenceToggle_e.swInputDimValOnCreate](#) to suppress the dialog box that allows the user to type the dimension value.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::AddDiameterDimension2 Method \(\)](#)
[IModelDoc2::IAddDiameterDimension2 Method \(\)](#)
[IModelDoc2::AddHorizontalDimension2 Method \(\)](#)
[IModelDoc2::IAddHorizontalDimension2 Method \(\)](#)
[IModelDoc2::AddRadialDimension2 Method \(\)](#)
[IModelDoc2::IAddRadialDimension2 Method \(\)](#)
[IModelDoc2::AddVerticalDimension2 Method \(\)](#)
[IModelDoc2::IAddVerticalDimension2 Method \(\)](#)
[IModelDocExtension::AddSymmetricDimension Method \(\)](#)
[IModelDocExtension::AddSpecificDimension Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddDisplayStateSpecificRenderMaterial Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddDisplayStateSpecificRenderMaterial Method (IModelDocExtension)

Adds the specified appearance to the specified display states in the active configuration and returns the IDs assigned to that appearance.

.NET Syntax

Visual Basic (Declaration)

```
Function AddDisplayStateSpecificRenderMaterial( _  
    ByVal PRenderMaterial As RenderMaterial, _  
    ByVal DisplayStateOption As System.Integer, _  
    ByVal DisplayStateNames As System.Object, _  
    ByRef PWMaterialId1 As System.Integer, _  
    ByRef PWMaterialId2 As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim PRenderMaterial As RenderMaterial  
Dim DisplayStateOption As System.Integer  
Dim DisplayStateNames As System.Object  
Dim PWMaterialId1 As System.Integer  
Dim PWMaterialId2 As System.Integer  
Dim value As System.Boolean  
  
value = instance.AddDisplayStateSpecificRenderMaterial(PRenderMaterial, DisplayStateOption,  
DisplayStateNames, PWMaterialId1, PWMaterialId2)
```

C#

```
System.bool AddDisplayStateSpecificRenderMaterial(  
    RenderMaterial PRenderMaterial,  
    System.int DisplayStateOption,  
    System.object DisplayStateNames,  
    out System.int PWMaterialId1,  
    out System.int PWMaterialId2  
)
```

C++/CLI

```

System.bool AddDisplayStateSpecificRenderMaterial(
& RenderMaterial^ PRenderMaterial,
& System.int DisplayStateOption,
& System.Object^ DisplayStateNames,
& [Out] System.int PWMaterialId1,
& [Out] System.int PWMaterialId2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PRenderMaterial
[Appearance](#) to add
DisplayStateOption
Display states as defined in [swDisplayStateOptions](#)
DisplayStateNames
Names of display states to which to add the appearance
PWMaterialId1
First ID of appearance
PWMaterialId2
Second ID of appearance

Return Value

True if appearance is added, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddDisplayStateSpecificRenderMaterial](#).

Example

[Add and Delete Appearances from Specific Display States \(C#\)](#)
[Add and Delete Appearances from Specific Display States \(VB.NET\)](#)
[Add and Delete Appearances from Specific Display States \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::DeleteDisplayStateSpecificRenderMaterial Method](#)
[IModelDocExtension::AddDisplayStateSpecificRenderMaterial Method](#)
[IModelDocExtension::DeleteDisplayStateSpecificRenderMaterial Method](#)
[IConfiguration::CreateDisplayState Method](#)
[IRenderMaterial Interface](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddFeatureMgrView3x64 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddFeatureMgrView3x64 Method (IModelDocExtension)

Adds the specified tab to the FeatureManager design tree view in x64 applications.

.NET Syntax

Visual Basic (Declaration)

```

Function AddFeatureMgrView3x64( _
    ByRef Bitmap As System.Long, _
    ByRef AppView As System.Long, _
    ByVal ToolTip As System.String, _
    ByVal WhichPane As System.Integer _
) As System.Boolean

```

Visual Basic (Usage)

```

Dim instance As IModelDocExtension
Dim Bitmap As System.Long

```



```
Dim AppView As System.Long
Dim ToolTip As System.String
Dim WhichPane As System.Integer
Dim value As System.Boolean
```

```
value = instance.AddFeatureMgrView3x64(Bitmap, AppView, ToolTip, WhichPane)
```

C#

```
System.bool AddFeatureMgrView3x64(
    ref System.long Bitmap,
    ref System.long AppView,
    System.string ToolTip,
    System.int WhichPane
)
```

C++/CLI

```
System.bool AddFeatureMgrView3x64(
    & System.int64% Bitmap,
    & System.int64% AppView,
    & System.String^ ToolTip,
    & System.int WhichPane
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bitmap

Pointer to the bitmap that you want to use for the FeatureManager design tree tab; the bitmap should be no larger than 16x18; this standard pointer is created by performing a New on Cbitmap

AppView

Pointer to an existing view that you want to use as your FeatureManager design tree view. If you do not have an existing view, then use [IModelViewMgr::CreateFeatureMgrView2](#), which returns a view handle

ToolTip

Tooltip string

WhichPane

Pane in which to add the view as defined in [swFeatMgrPane_e](#)

Return Value

True if the FeatureManager design tree view tab is added successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddFeatureMgrView3x64](#).

Remarks

See the Remarks in [IModelDoc2::AddFeatureMgrView3](#) for more information about using this x64 method.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::ActivateFeatureMgrViewx64 Method \(\)](#)

[IModelDocExtension::DeActivateFeatureMgrViewx64 Method \(\)](#)

Availability

SOLIDWORKS 2026 SP01, Revision Number 34.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddOrdinateDimension Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddOrdinateDimension Method (IModelDocExtension)

Inserts an ordinate dimension.

.NET Syntax

```
Function AddOrdinateDimension( _
    ByVal DimType As System.Integer, _
    ByVal LocX As System.Double, _
    ByVal LocY As System.Double, _
    ByVal LocZ As System.Double _
) As System.Integer
```



```
Dim instance As IModelDocExtension
Dim DimType As System.Integer
Dim LocX As System.Double
Dim LocY As System.Double
Dim LocZ As System.Double
Dim value As System.Integer

value = instance.AddOrdinateDimension(DimType, LocX, LocY, LocZ)
```

```
System.int AddOrdinateDimension(
    System.int DimType,
    System.double LocX,
    System.double LocY,
    System.double LocZ
)
```

```
System.int AddOrdinateDimension(
    & System.int DimType,
    & System.double LocX,
    & System.double LocY,
    & System.double LocZ
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DimType

Dimension type as defined in [swAddOrdinateDims_e](#)

LocX

X location for the dimension

LocY

Y location for the dimension

LocZ

Z location for the dimension

Return Value

Error as defined by [swCreateOrdDimError_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddOrdinateDimension](#).

Example

[Display Grid Bubble \(VBA\)](#)
[Create Ordinate Dimensions \(C#\)](#)
[Create Ordinate Dimensions \(VB.NET\)](#)
[Create Ordinate Dimensions \(VBA\)](#)

Remarks

Before using this method, select the base entity to act as the datum point for the ordinate dimension and any additional entities to include in the group of ordinate dimensions.

Selections made immediately after calling this method continue to add ordinate dimensions to the group of ordinate dimensions. When you finish adding ordinate dimensions to the group, use [IModelDoc2::SetPickMode](#) to return to the default selection mode.

Use [IModelDoc2::EditOrdinate](#) to add ordinate dimensions to an existing group of ordinate dimensions.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IDisplayDimension Interface](#)
[IModelDocExtension::JogDimension Method](#)
[IModelDoc2::AlignOrdinate Method](#)
[IModelDoc2::ReattachOrdinate Method](#)
[IDrawingDoc::AlignOrdinate Method](#)
[IDrawingDoc::CreateOrdinateDim4 Method](#)
[IDrawingDoc::EditOrdinate Method](#)
[IDrawingDoc::InsertHorizontalOrdinate Method](#)
[IDrawingDoc::InsertVerticalOrdinate Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddOrUpdateSearchData Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddOrUpdateSearchData Method (IModelDocExtension)

Adds or updates the SOLIDWORKS Search, third-party, application keyword and value to the model document.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function AddOrUpdateSearchData( _  
    ByVal AppName As System.String, _  
    ByVal AppKeyword As System.String, _  
    ByVal AppValue As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim AppName As System.String  
Dim AppKeyword As System.String  
Dim AppValue As System.String  
Dim value As System.Boolean  
  
value = instance.AddOrUpdateSearchData(AppName, AppKeyword, AppValue)
```

C#

```
System.bool AddOrUpdateSearchData(  
    System.string AppName,  
    System.string AppKeyword,  
    System.string AppValue  
)
```

C++/CLI

```
System.bool AddOrUpdateSearchData(  
& System.String^ AppName,  
& System.String^ AppKeyword,  
& System.String^ AppValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AppName
Third-party application name
AppKeyword
Third-party application keyword
AppValue
Value for AppKeyword

Return Value

True if the SOLIDWORKS Search, third-party, application keyword and value are added to the model document, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddOrUpdateSearchData](#).

Example

[Add Third-party Application Keywords to SOLIDWORKS Search and Model \(VBA\)](#)

Remarks

After calling this method and after the keyword has been indexed in SOLIDWORKS Search, type the third-party application name, keyword, or keyword value in the SOLIDWORKS Search box and press **Enter**. Any documents assigned any of these three strings will appear on the **Search** tab in the Task Pane.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::DeleteSearchData Method](#)
[IModelDocExtension::GetSearchData Method](#)
[IModelDocExtension::GetSearchDataCount Method](#)
[IModelDocExtension::IGetSearchData Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddPathLengthDim Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddPathLengthDim Method (IModelDocExtension)

Inserts a path length dimension at the specified coordinates for a selected path.

.NET Syntax

Visual Basic (Declaration)

```
Function AddPathLengthDim( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.AddPathLengthDim(X, Y, Z)
```

C#

```
System.object AddPathLengthDim(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI



```
System.Object^ AddPathLengthDim(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X coordinate of display dimension
Y
Y coordinate of display dimension
Z
Z coordinate of display dimension

Return Value

[IDisplayDimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddPathLengthDim](#).

Example

[Create Path Length Dimension \(VBA\)](#)
[Create Path Length Dimension \(VB.NET\)](#)
[Create Path Length Dimension \(C#\)](#)

Remarks

Before calling this method to create a path length dimension:

If a path...	Then...
Exists	Call IModelDocExtension::SelectByID2 to select one sketch segment on the existing path.
Does not exist	1. Call IModelDocExtension::SelectByID2 to select two or more sketch segments that are end-to-end coincident and form a single chain. 2. Call SketchManager::MakeSketchChain to create a path with the selected sketch segments. 3. Call IModelDocExtension::SelectByID2 to select one sketch segment on the path.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic](#).

AddRenderMaterial Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddRenderMaterial Method (IModelDocExtension)

Not supported in SOLIDWORKS 2011 and later. Superseded by [IModelDocExtension::AddDisplayStateSpecificRenderMaterial](#) and [IModelDocExtension::IAddDisplayStateSpecificRenderMaterial](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddRenderMaterial(  
    ByVal PRenderMaterial As RenderMaterial, _  
    ByRef PwMaterialId As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim PRenderMaterial As RenderMaterial
```



```
Dim PwMaterialId As System.Integer
Dim value As System.Boolean

value = instance.AddRenderMaterial(PRenderMaterial, PwMaterialId)
```

C#

```
System.bool AddRenderMaterial(
    RenderMaterial PRenderMaterial,
    out System.int PwMaterialId
)
```

C++/CLI

```
System.bool AddRenderMaterial(
    & RenderMaterial^ PRenderMaterial,
    & [Out] System.int PwMaterialId
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PRenderMaterial
[Appearance](#) to add to the model
PwMaterialId
 Appearance ID

Return Value

True if the appearance is added to the model, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddRenderMaterial](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddSpecificDimension Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddSpecificDimension Method (IModelDocExtension)

Creates the specified display dimension at the specified location for selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Function AddSpecificDimension( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double, _
    ByVal DimensionType As System.Integer, _
    ByRef Error As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
```

```

Dim DimensionType As System.Integer
Dim Error As System.Integer
Dim value As System.Object

value = instance.AddSpecificDimension(X, Y, Z, DimensionType, Error)

```

C#

```

System.Object AddSpecificDimension(
    System.double X,
    System.double Y,
    System.double Z,
    System.int DimensionType,
    out System.int Error
)

```

C++/CLI

```

System.Object^ AddSpecificDimension(
    & System.double X,
    & System.double Y,
    & System.double Z,
    & System.int DimensionType,
    & [Out] System.int Error
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X coordinate of display dimension text

Y
Y coordinate of display dimension text

Z
Z coordinate of display dimension text

DimensionType
Type of dimension to add as defined in [swDimensionType_e](#) (see **Remarks**)

Error
Result status as defined in [swAddSpecificDimension_e](#)

Return Value

[IDisplayDimension](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddSpecificDimension](#).

Example

[Create Specific Dimension in a Sketch \(VBA\)](#)
[Create Specific Dimension in a Sketch \(VB.NET\)](#)
[Create Specific Dimension in a Sketch \(C#\)](#)

Remarks

This method adds specific display dimensions to sketches and drawings. It supports the following from `swDimensionType_e`:

- `swAngularDimension`
- `swChamferDimension`
- `swDiameterDimension`
- `swDiametricLinearDimension`
- `swHorLinearDimension`
- `swOrdinateDimension`
- `swRadialDimension`
- `swRadialLinearDimension`
- `swVertLinearDimension`

If you specify `DimensionType` with `swDimensionType_e`:

- `swDiametricLinearDimension`, then this method creates a double-distance linear dimension for a selected construction/center line, circle/arc edge/center. After creation, use [IDisplayDimension::Diametric](#) to display the single-distance linear dimension.
- `swRadialLinearDimension`, then this method creates a single-distance linear dimension for a selected construction/center line, circle/arc edge/center. After creation, use [IDisplayDimension::Diametric](#) to display the double-distance linear dimension.

Before calling this method, you must select the entities to dimension.

For example, to create an angular dimension between two intersecting lines in a drawing:

1. Call [IModelDocExtension::SelectByRay](#), specifying the X, Y, and Z coordinates of a ray of the angle you want to dimension.
2. Call [IModelDocExtension::SelectByRay](#), specifying the X, Y, and Z coordinates of the other ray of the angle you want to dimension.
3. Call this method, specifying X, Y, and Z with the location coordinates of the display dimension text and specifying `DimensionType` with `swDimensionType_e.swAngularDimension`.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::AddDimension Method \(\)](#)
[IModelDocExtension::AddSymmetricDimension Method \(\)](#)
[IModelDoc2::AddDiameterDimension2 Method \(\)](#)
[IModelDoc2::AddHorizontalDimension2 Method \(\)](#)
[IModelDoc2::AddRadialDimension2 Method \(\)](#)
[IModelDoc2::AddVerticalDimension2 Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddSymmetricDimension Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AddSymmetricDimension Method (IModelDocExtension)

Creates a full symmetrical angular dimension at the specified location for the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Function AddSymmetricDimension( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.AddSymmetricDimension(X, Y, Z)
```

C#

```
System.object AddSymmetricDimension(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.Object^ AddSymmetricDimension(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X coordinate of the symmetrical angular dimension
Y
Y coordinate of the symmetrical angular dimension
Z
Z coordinate of the symmetrical angular dimension

Return Value

Symmetrical angular [dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AddSymmetricDimension](#).

Example

[Create Full Symmetrical Angular Dimension \(C#\)](#)
[Create Full Symmetrical Angular Dimension \(VB.NET\)](#)
[Create Full Symmetrical Angular Dimension \(VBA\)](#)

Remarks

This type of dimensioning is helpful when you create a sketch for revolved geometry that requires a full angular dimension.

Before calling this method, you might want to call [ISldWorks::SetUserPreferenceToggle](#) with [swUserPreferenceToggle e.swInputDimValOnCreate](#) to suppress the dialog box that allows the user to enter the dimension value.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::AddDimension Method \(\)](#)
[IModelDoc2::AddDiameterDimension2 Method \(\)](#)
[IModelDoc2::AddDiameterDimension2 Method \(\)](#)
[IModelDoc2::AddHorizontalDimension2 Method \(\)](#)
[IModelDoc2::AddHorizontalDimension2 Method \(\)](#)
[IModelDoc2::AddRadialDimension2 Method \(\)](#)
[IModelDoc2::AddRadialDimension2 Method \(\)](#)
[IModelDoc2::AddVerticalDimension2 Method \(\)](#)
[IModelDoc2::AddVerticalDimension2 Method \(\)](#)
[IModelDocExtension::AddSpecificDimension Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AlignDimensions Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AlignDimensions Method (IModelDocExtension)

Aligns the selected dimensions in drawing documents.

.NET Syntax

Visual Basic (Declaration)

```
Function AlignDimensions( _  
    ByVal AlignType As System.Integer, _  
    ByVal SpaceValue As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim AlignType As System.Integer  
Dim SpaceValue As System.Double  
Dim value As System.Boolean
```

```
value = instance.AlignDimensions(AlignType, SpaceValue)
```

C#

```
System.bool AlignDimensions(  
    System.int AlignType,  
    System.double SpaceValue  
)
```

C++/CLI

```
System.bool AlignDimensions(  
& System.int AlignType,  
& System.double SpaceValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AlignType

Type of alignment as defined by [swAlignDimensionType_e](#)

SpaceValue

Distance between dimensions

Return Value

True if the dimensions are aligned, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AlignDimensions](#).

Example

[Auto-arrange Dimensions \(C#\)](#)
[Auto-arrange Dimensions \(VB.NET\)](#)
[Auto-arrange Dimensions \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::AlignParallelDimensions Method](#)
[IModelDoc2::AlignOrdinate Method](#)
[IModelDoc2::BreakDimensionAlignment Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AlignRunningDimension Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AlignRunningDimension Method (IModelDocExtension)

Aligns the extension lines of all angular dimensions to be the same distance from the center as the baseline dimension (0°) in the set of angular running dimensions.

.NET Syntax

Visual Basic (Declaration)

```
Sub AlignRunningDimension()
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
```

```
instance.AlignRunningDimension()
```

C#

```
void AlignRunningDimension()
```

C++/CLI

```
void AlignRunningDimension();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AlignRunningDimension](#).

Example

[Insert Angular Running Dimension \(VBA\)](#)
[Insert Angular Running Dimension \(VB.NET\)](#)
[Insert Angular Running Dimension \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::ReJogRunningDimension Method \(\)](#)
[IModelDocExtension::AddAngularRunningDim Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AnnotationViewCount Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AnnotationViewCount Property (IModelDocExtension)

Gets the number of annotation views in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property AnnotationViewCount As System.Integer

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
```

```
Dim value As System.Integer
```

```
value = instance.AnnotationViewCount
```

C#

```
System.int AnnotationViewCount {get;}
```

C++/CLI

```
property System.int AnnotationViewCount {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Number of annotation views in this model document

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AnnotationViewCount](#).

Example

[Move Annotations to Notes Area Annotation View \(C#\)](#)
[Move Annotations to Notes Area Annotation View \(VB.NET\)](#)
[Move Annotations to Notes Area Annotation View \(VBA\)](#)

Remarks

Call this method before calling [IModelDocExtension::IGetAnnotationViews](#) to get the size of the array for that method.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetAnnotationViews Method](#)
[IModelDocExtension::AnnotationViews Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AnnotationViews Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AnnotationViews Property (IModelDocExtension)

Gets the annotation views in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property AnnotationViews As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Object  
  
value = instance.AnnotationViews
```

C#

```
System.Object AnnotationViews {get;}
```

C++/CLI

```
property System.Object^ AnnotationViews {  
    System.Object^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [annotation views](#) in this part or assembly document

Visual Basic for Applications (VBA) Syntax

See [IModelDocExtension::AnnotationViews](#).

Example

[Move Annotations to Notes Area Annotation View \(C#\)](#)
[Move Annotations to Notes Area Annotation View \(VB.NET\)](#)
[Move Annotations to Notes Area Annotation View \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetAnnotationViews Method](#)
[IModelDocExtension::AnnotationViewCount Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ApplyFormatPainterToAll Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : [ApplyFormatPainterToAll Method \(IModelDocExtension\)](#)

Applies Format Painter to all dimensions and annotations in the active document.

.NET Syntax

Visual Basic (Declaration)

```
Sub ApplyFormatPainterToAll()
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
instance.ApplyFormatPainterToAll()
```

C#

```
void ApplyFormatPainterToAll()
```

C++/CLI

```
void ApplyFormatPainterToAll();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ApplyFormatPainterToAll](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::StartFormatPainter Method](#)
[IModelDocExtension::StopFormatPainter Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ApplyTexture Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : [ApplyTexture Method \(IModelDocExtension\)](#)

Obsolete. Superseded by [IModelDocExtension::SetTexture](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ApplyTexture( _  
    ByVal Scale As System.Integer, _  
    ByVal Angle As System.Double, _  
    ByVal TextureFilename As System.String, _  
    ByVal BlendColor As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Scale As System.Integer  
Dim Angle As System.Double  
Dim TextureFilename As System.String  
Dim BlendColor As System.Boolean
```

```
Dim value As System.Boolean
```

```
value = instance.ApplyTexture(Scale, Angle, TextureFilename, BlendColor)
```

C#

```
System.bool ApplyTexture(  
    System.int Scale,  
    System.double Angle,  
    System.string TextureFilename,  
    System.bool BlendColor  
)
```

C++/CLI

```
System.bool ApplyTexture(  
& System.int Scale,  
& System.double Angle,  
& System.String^ TextureFilename,  
& System.bool BlendColor  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Scale

Angle

TextureFilename

BlendColor

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ApplyTexture](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AppPageSetup Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : AppPageSetup Property (IModelDocExtension)

Gets the SOLIDWORKS Design application page setup interface for this document.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property AppPageSetup As PageSetup
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As PageSetup
```

```
value = instance.AppPageSetup
```

C#

```
PageSetup AppPageSetup {get;}
```

C++/CLI

```
property PageSetup^ AppPageSetup {  
    PageSetup^ get();
```

}

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Page setup](#) object for SOLIDWORKS Design

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::AppPageSetup](#).

Example

[Get Whether Draft Edges Scaled in Printed Drawing \(C#\)](#)
[Get Whether Draft Edges Scaled in Printed Drawing \(VB.NET\)](#)
[Get Whether Draft Edges Scaled in Printed Drawing \(VBA\)](#)

Remarks

The page setup object used when printing depends on a document setting, which indicates whether to use the application page setup, the page setup for this document.

To...	Use...
Get or set a document setting	IModelDocExtension::UsePageSetup
Get the document page setup	IModelDoc2::PageSetup or IModelDoc2::IPageSetup
Get the sheet page setup	ISheet::PageSetup or ISheet::IPageSetup

Although the object that this property returns indicates that it contains application-level values, some of the values that it contains depend on the type of document. Therefore, this API is on a document interface, and you should retrieve a different one for each different document that you print.

This property is read only.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

BreakAllExternalFileReferences2 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : BreakAllExternalFileReferences2 Method (IModelDocExtension)

Breaks all external references and allows you to insert the features of the original part, or parts, if the external references are broken.

.NET Syntax

Visual Basic (Declaration)

```
Sub BreakAllExternalFileReferences2(  
    ByVal InsertFeatures As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim InsertFeatures As System.Boolean  
  
instance.BreakAllExternalFileReferences2(InsertFeatures)
```

C#

```
void BreakAllExternalFileReferences2(  
    System.bool InsertFeatures  
)
```

C++/CLI

```
void BreakAllExternalFileReferences2(  
&    System.bool InsertFeatures  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InsertFeatures

True to insert the features of the original parts if the external references are broken, false to not

Return Value

See [ModelDocExtension::BreakAllExternalFileReferences2](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::BreakAllExternalFileReferences2](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::GetExternalReferenceName Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferencesCount Method](#)
[IModelDoc2::LockAllExternalReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferencesCount Method](#)
[IModelDocExtension::UpdateExternalFileReferences Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

Capture3DView Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : Capture3DView Method (IModelDocExtension)

Captures the 3D View of this part or assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function Capture3DView() As View3D
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As View3D  
  
value = instance.Capture3DView()
```

C#

```
View3D Capture3DView()
```

C++/CLI

```
View3D^ Capture3DView();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IView3D](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::Capture3DView](#).

Example

[Capture 3D View \(VBA\)](#)
[Capture 3D View \(VB.NET\)](#)
[Capture 3D View \(C#\)](#)

Remarks

This method requires that the SOLIDWORKS MBD add-in be loaded. ([ISldWorks::LoadAddIn](#))

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::Get3DView Method \(\)](#)
[IModelDocExtension::Get3DViews Method \(\)](#)
[IModelDocExtension::Refresh3DViews Method \(\)](#)
[IModelDocExtension::Get3DViewCount Method \(\)](#)
[IModelDocExtension::Get3DViewNames Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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ChangeSketchPlane Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ChangeSketchPlane Method (IModelDocExtension)

Changes the plane used by a sketch by moving the selected sketch to the selected plane in the specified configurations.

.NET Syntax

Visual Basic (Declaration)

```
Function ChangeSketchPlane( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_names As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Config_opt As System.Integer  
Dim Config_names As System.Object  
Dim value As System.Boolean
```

```
value = instance.ChangeSketchPlane(Config_opt, Config_names)
```

C#

```
System.bool ChangeSketchPlane(  
    System.int Config\_opt,  
    System.object Config\_names  
)
```

C++/CLI

```
System.bool ChangeSketchPlane(  
& System.int Config\_opt,
```

```
& System.Object^ Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt
Configurations as defined by [swInConfigurationOpts_e](#)
Config_names
Array of configuration names

Return Value

True if the operation succeeds, false if it fails

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ChangeSketchPlane](#).

Example

[Change Plane of Sketch \(VBA\)](#)
[Change Plane of Sketch \(VB.NET\)](#)
[Change Plane of Sketch \(C#\)](#)

Remarks

Every sketch is associated with a plane (for example, a reference plane or a planar face). You must preselect the sketch and the new plane or face before using this method.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::IChangeSketchPlane Method](#)

Availability

SOLIDWORKS 2008 SP04, Revision Number 16.4